1 Attribute types

Select the attribute types for the given table:

rank	title	year	runtime (min)	budget ($)	avg rating
1	The Shawshank Redemption	1994	142	25 000 000	9.3
2	The Godfather	1972	175	6 000 000	9.2
3	The Godfather: Part II	1974	202	13 000 000	9.0
4	The Dark Knight	2008	152	185 000 000	9.0
5	12 Angry Men	1957	96	350 000	8.9

- rank: Select alternative (ordinal, quantitative, categorical)
- title Select alternative (categorical, quantitative, ordinal)
- budget: Select alternative (ordinal, quantitative, categorical)
- avg rating: Select alternative (quantitative, categorical, ordinal)

Maximum marks: 4

2 Scatterplots Matrices and Parallel Coordinates

Scatterplots Matrices and Parallel Coordinates support different tasks.

State 4 tasks/findings that can be accomplished by using a Scatterplot Matrix in general!

State 1 tasks/finding that can be accomplished better in Parallel Coordinates in general!

Maximum marks: 10

3 Comparison of Star Plots and Parallel Coordinates

Think about Star plots and Parallel Sets!

State two things they have in common in layout.

State one thing they differ in layout.

State one advantage of a Starplot over a Parallel Coordinates Plot.

State one advantage of a Parallel Coordinates Plot over a Starplot.

Maximum marks: 10

4 Visual Channels

Think about visual channels

Please order the following visual channels for a continuous attribute by effectiveness starting with the most effective channel {Length; Position on unaligned scale; Volume;Color saturation; Area }

Please explain what popout effects are!

Please name three example channels where popout effects exist!

Maximum marks: 10

5 Scalability of time-oriented visualizations

State examples of time-oriented visualizations that work well for the numbers of time series mentioned below. A line chart, a bar chart or a scatterplot is not allowed. Each time-oriented visualization can be mentioned only once.

5 time series

20 time series

100 time series

Maximum marks: 6

6 Tree Visualizations and inner Relations

Match these typical tree drawing techniques to their properties.

Highlight for each plot type its way to express the parent-child relationship

	Adjacency	Containment	Line
Sunburst Diagram	○	○	○
Node-Link-Diagrams	○	○	○
Treemaps	○	○	○

Maximum marks: 6

7 Trees and tasks

Which tasks that can be fulfilled well with which plot type. Each option can only be used once.

	spot outliers in leaf nodes	following paths	sum up values of children
Icycle Layout	○	○	○
Treemaps	○	○	○
Node-Link Diagrams	○	○	○

Maximum marks: 6

8 Daily temperature

State advantages and disadvantages of the following chart types for visualizing the maximum daily temperature over one year in 10 different cities.

(a) line graph with grid

(b) braided graph

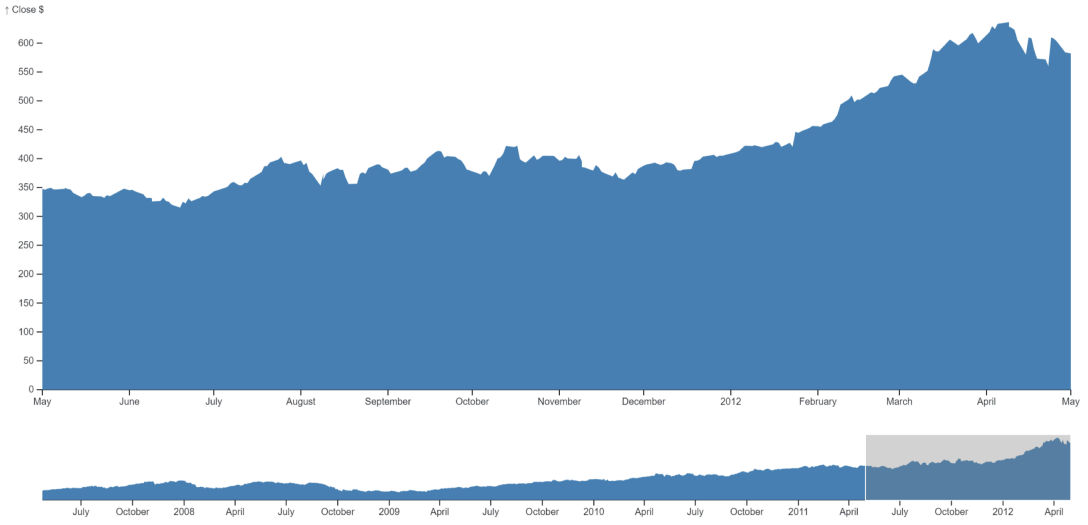
(c) stacked area chart (stream graph)

(a) small multiples

(e) horizon graphs

Maximum marks: 10

9 InfoVis Techniques



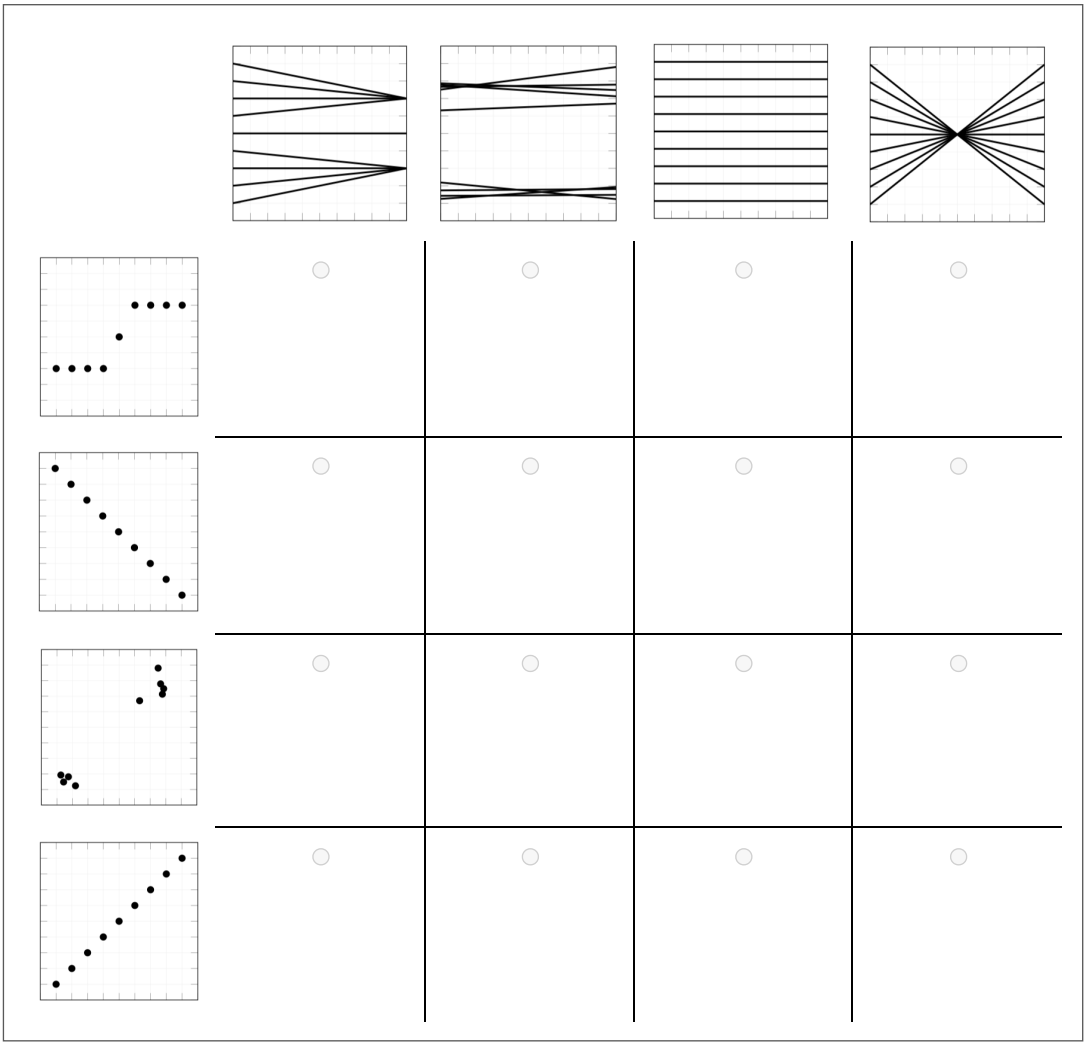
What kind of technique is being used here:

- Semantic Zooming
- Overview+Detail
- Focus andContext
- Detail on Demand

Maximum marks: 2

10 Pattern Matching

Match the Scatterplots and Parallel Coordinates that visualize the same data



Maximum marks: 8

11 Graph Visualization

Which kind of Visualization are suited for Graphs?

Graphs

- Enclosure
- Node-Link-Diagrams
- Adjacency-Diagrams
- Adjacency-Matrix

Maximum marks: 4

12 Tree Visualizations

Which kind of Visualization are suited for Trees?

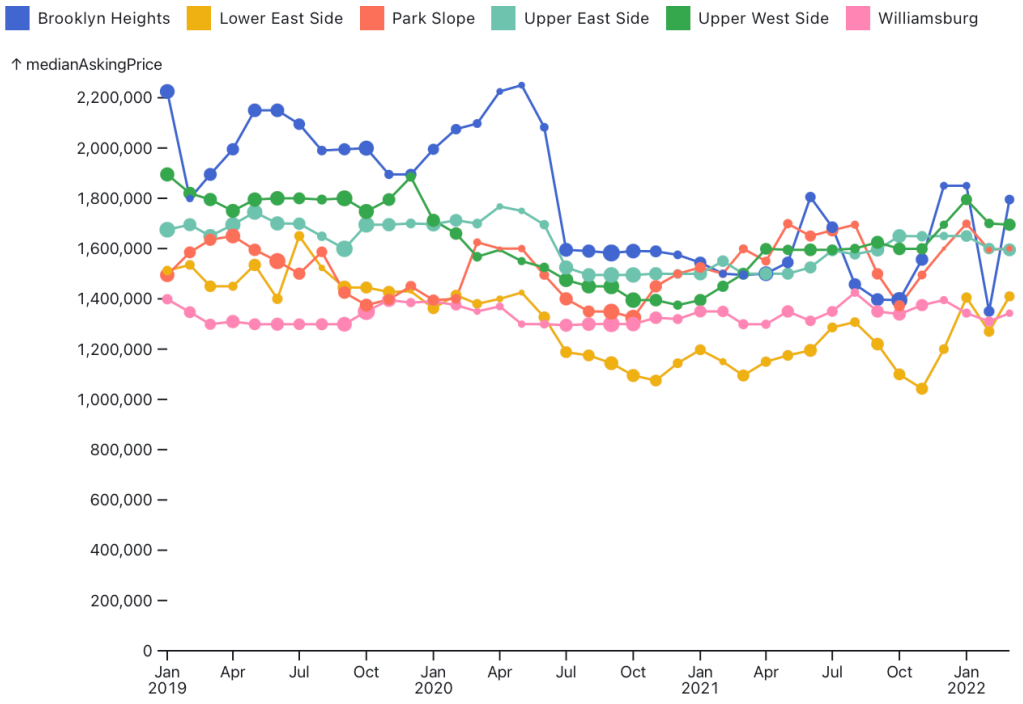
Trees

- [ ] Adjacency-Matrix
- [ ] Enclosure
- [ ] Adjacency-Diagrams
- [ ] Node-Link-Diagrams

Maximum marks: 4

13 Plot 1

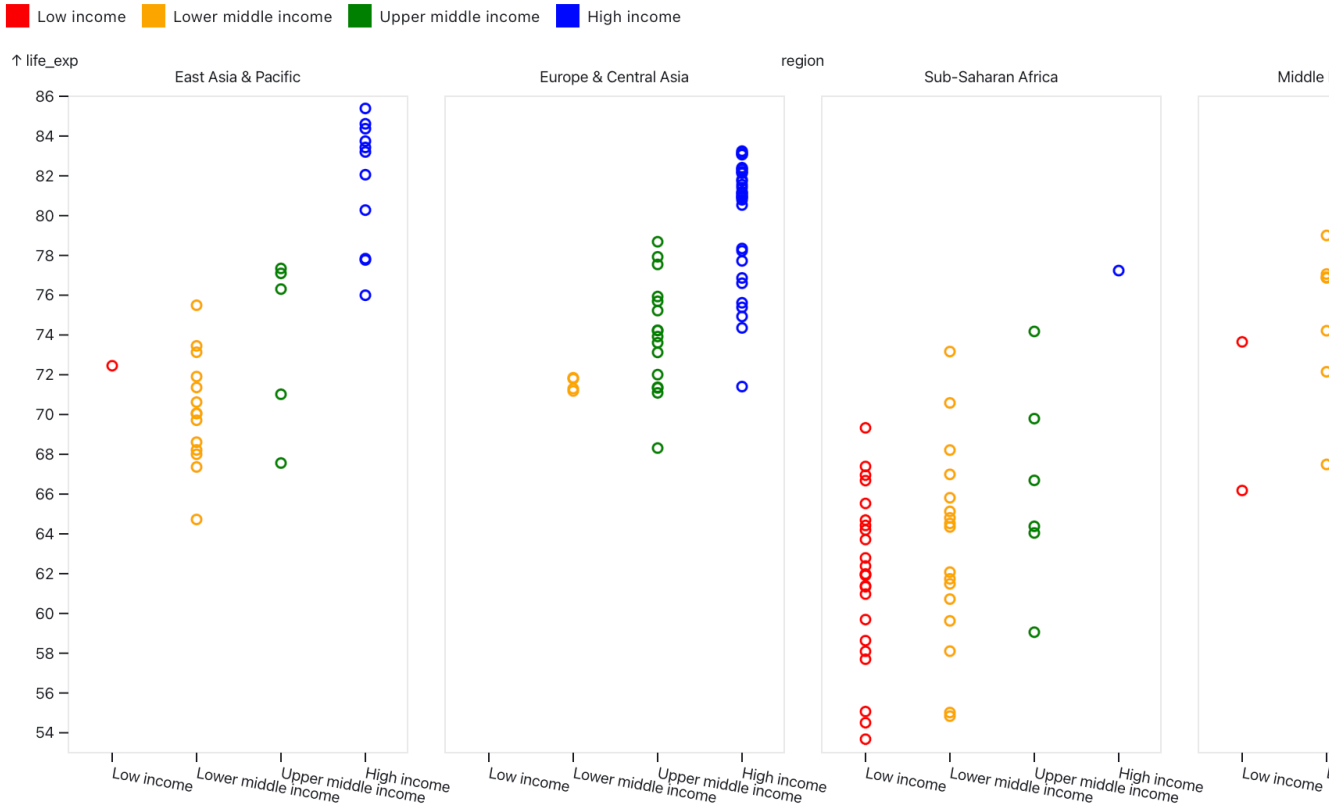
Recreate the following chart with Observable Plot Syntax for the dataset "nyData". The x axis should show "date" and y the "medianAskingPrice". It is colorcoded by "neighborhood" and radius of the circles should by mapped to "priceCutShare".



Fill in your answer here

Maximum marks: 15

14 Plot 2



Recreate the following chart with Observable Plot Syntax for the dataset "worldbank". The x axis should show "income_group" and y the "life_exp". It is colorcoded by "income_group" and faceted by "region".
Add the following things:

- A legend
- A color scale the matches the actual colors used in the figure with domain values as shown on x.

Further, adjust vertical axis according to the shown values.

Fill in your Plot Code

Maximum marks: 15